

Course Grading Guidelines

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Classification and Grading

There is no exam. Your grade will be based on your independent work. There are three outputs used for grading.

1. **Journal Club Presentation:** 20% of the final grade
2. **Capstone Project Written Report:** 50% of the final grade
3. **Poster Presentation:** 30% of the final grade

1 Journal Club

A journal club is a group who meet to critically evaluate articles in the academic literature. Journal clubs:

- Encourages critical thinking.
- Helps in understanding the studied concepts and methods.
- Helps keep up with the latest research.
- Improves communication and presentation skills.
- Presentation simulates experience of a scientist.

The main purpose in this course is to explore how methods we learned are applied in current research.

1.1 Your Task for Journal Club

- You will read a research paper related to the topic of your capstone project.
- Your task is to read the paper and prepare a 15-minute presentation, followed by a 5-minute discussion, to explain the paper.
- The presentation dates will be arranged individually.

1.2 Journal Club Presentation Content

The Journal Club presentation will cover:

- **Introduction:** Provide a High-level Overview on the background of the research topic and the main objectives of the paper.
- **Methods:** Give overview of the methodology used in the paper.
- **Results:** Highlight the key results and conclusions of the paper.
- **Discussion:** Discuss the practical implications of the research and its applications in real-world situations. Discuss any limitations of the paper.

- **Conclusions:** Summarize the main contributions and overall findings of **the paper**. Reiterate its key takeaways and their broader implications.
- **Q&A:** Answer questions from the audience (lecturer).

1.3 Grading Criteria for Journal Club Presentation

Presentation Design and Delivery (50%)

- **Design:** The presentation is well-organized and easy to follow.
- **Timing:** The presenter adhered to the 15-minute presentation format.
- **Delivery:** Clear and confident speech with appropriate pacing.
- **Questions:** Convincing response to questions.

Content of the presentation (50%)

- **Introduction:** The introduction provides a high-level overview of the topic: explains the research motivation and the research question.
- **Methods:** Explains the experimental approach used by the authors. Unfamiliar methods are explained without going into excessive detail.
- **Results:** The key results from the paper are explained.
- **Critical evaluation:** For example critique of methods, ideas for improvement, significance of the results.

2 Capstone Project

The Capstone project is your independent project.

- You will work on an independent project that builds on and further extends course content.
- Output of your work will be a written report (formatted as an academic article), a poster, and code.

2.1 Capstone Project Report Content

The Capstone Project written report will cover:

- **Introduction:** Provide a High-level Overview on the background of the research topic and the main objectives of **your project**.
- **Methods:** Give overview of the methodology used in **your project**.
- **Results:** Highlight the key results and conclusions of **your project**.
- **Discussion:** Discuss the practical implications of **your project's findings** and their applications in real-world situations. Discuss any limitations of **your project**.
- **Conclusions:** Summarize the main contributions and overall findings of **your project**. Reiterate its key takeaways and their broader implications.

2.2 Grading Criteria for Capstone Project

Report Quality and Structure (50%)

- **Formatting:** The report is well-organized, logically structured, and follows the conventions of a scientific article.
- **Writing Clarity:** The text is clear, professional, and effectively communicates complex concepts.
- **Plots and Figures:** Clear graphical presentation of simulation results that address project objectives and show model behavior. All figures are properly labeled and referenced in the text.

Model Implementation and Analysis (50%)

- **Correctness:** Correct function of fundamental models.
- **Understanding:** Demonstrated understanding of core concepts as articulated in the report.
- **Extension beyond core:** Successful and meaningful implementation of additional features (if the task specifies them).
- **Discussion:** Insightful written analysis on how parameters/initial conditions affect outcomes. Identification and thorough explanation of behaviors.

3 Poster Presentation

A poster presentation is a visual communication tool commonly used at academic conferences. It combines text and graphics to present your project in a way that is visually accessible and encourages one-on-one discussion. Poster sessions:

- Develop skills in synthesizing complex information into a concise visual format.
- Improve the ability to engage in interactive, informal professional discussions.
- Simulate real-world networking and conference experiences.

3.1 Your Task for the Poster

- You will design a scientific poster based on your Capstone Project.
- You should be prepared to give a short (e.g., 3-minute) "elevator pitch" summarizing your work to the audience.
- You will stand by your poster to answer questions and discuss your findings in an informal setting.

3.2 Poster Content and Layout

A successful poster should be visually engaging and easy to read from a distance. It typically includes:

- **Header:** Title, author(s), and affiliations.
- **Introduction/Objectives:** Briefly state the problem, motivation, and the goals of your project.
- **Methods:** Visual overview of your approach (flowcharts or diagrams are highly encouraged).
- **Results:** Key findings presented primarily through clear, well-labeled plots, figures, or tables. Keep text to a minimum.
- **Conclusions:** Bullet points summarizing the main takeaways and future directions.

3.3 Grading Criteria for Poster Presentation

Visual Design and Layout (50%)

- **Readability:** Text is legible from a distance; clear font choices and appropriate sizing.
- **Visuals:** Effective use of charts, graphs, and diagrams instead of dense blocks of text.
- **Organization:** Logical flow of information that guides the reader's eye naturally from introduction to conclusion.

Content and Interactive Discussion (50%)

- **Clarity of Message:** The core objective and main findings are immediately clear.
- **Elevator Pitch:** Ability to concisely summarize the poster in a few minutes without simply reading directly from it.
- **Q&A Engagement:** Confident, accurate, and engaging responses to questions posed during the poster session.